5.4 Hypothesis tests on identification results

To distinguish reproducible accuracy differences from realization-dependent fluctuations, the final estimates were compared across the 30 independent Monte Carlo records. The outcome was the relative parameter error expressed as a percentage,

$$E_{a,r} = 100\frac{\|\hat{\theta}_{a,r} - \theta\|_2}{\|\theta\|_2}, \tag{78}$$

where a denotes a method and r a realization. Each BI-HMC observation is the error of one physical posterior-mean estimate for one record; posterior draws are not treated as independent experimental replications.

Distributional diagnostics and choice of tests. Shapiro–Wilk tests [4] assessed the normality of each method's 30 final errors, retaining all finite observations, including extreme values. Their Holm-adjusted results in Table 9 reject normality for 32 of the 36 testable method–case distributions. The exceptions are LHAM-3-TIK for Example 1/PL, and WS-GGI, GHAM-2-I, and LHAM-1-TIK for CSTR/PL; non-rejection with 30 observations does not establish a normal population. The non-Gaussian distributions motivate comparisons that do not require Gaussian errors. However, marginal normality tests do not verify pairing or independence, and do not test the normality of paired differences required for exact paired t inference. The tests below were selected to match the sampling design and the intended accuracy comparison, rather than selected case-by-case according to the Shapiro–Wilk decisions.

For methods fitted to the same records, we used a two-sided paired exact sign test [1]. For each pair (a, b), let W_{ab} and L_{ab} count the realizations with $E_{a,r} < E_{b,r}$ and $E_{a,r} > E_{b,r}$, respectively. Numerical ties were excluded, using $\tau_r = 10\,\mathrm{eps}(\max\{1, E_{a,r}, E_{b,r}\})$ as the equality tolerance, where $\mathrm{eps}(x)$ is floating-point spacing at x. Under equal win probabilities, $W_{ab} \mid n_{ab} \sim \mathrm{Binomial}(n_{ab}, 1/2)$, with $n_{ab} = W_{ab} + L_{ab}$. The exact two-sided tail probability tests whether one method produces smaller errors more frequently. This test preserves the common-record pairing and requires neither Gaussian errors nor the symmetry of paired differences assumed by the Wilcoxon signed-rank test. Its conclusion concerns win frequency, not equality of arithmetic means.

The PL presentation reuses the BI-HMC reference computed in Python on separate records from the MATLAB PL runs. Matching run numbers or nominal seeds therefore does not provide a valid pairing for these contrasts. They were evaluated using two-sided independent Wilcoxon rank-sum (Mann–Whitney) tests [3], using the tie-corrected normal approximation with continuity correction. The null is equality of the two error distributions. Direction is summarized by

$$\widehat{A}_{ab} = \frac{1}{n_a n_b}\sum_{i=1}^{n_a}\sum_{j=1}^{n_b}\left[\mathbf{1}\{E_{a,i} < E_{b,j}\} + \tfrac{1}{2}\mathbf{1}\{E_{a,i} = E_{b,j}\}\right], \tag{79}$$

where $\widehat{A}_{ab} > 1/2$ favors a. This rank-based distributional comparison does not require normal samples; without a common-shape location model it should not be interpreted solely as a test of medians or as proof of lower errors at every quantile. The $n_a n_b$ comparisons in (79) are dependent and were not used as independent binomial trials. The same saved BI-HMC reference is displayed under both parametrizations and does not constitute an additional set of independent Bayesian experiments.

All pairwise hypotheses were tested once, two-sided, and corrected jointly within each benchmark–parametrization case using Holm's procedure at $\alpha = 0.05$ [2]. Paired and independent tests belong to the same family when both occur in a case. The separately reported Shapiro–Wilk tests form their own diagnostic families. Unavailable comparisons produce no rejection; no finite extreme errors are discarded. Where legacy PL files lack per-run hashes, pairing among the PL methods follows the saved realization columns, as prescribed by the experiment generator. All testable Monte Carlo methods supplied 30 valid final estimates.

Statistical evidence across benchmarks. Tables 10 and 11 summarize the supported pairwise advantages. Figures 19– 22 report every comparison and adjusted p-value. Example 1/FSM yields no significant pairwise difference, including comparisons with BI-HMC, despite its lower mean error in the descriptive table. For Example 1/PL, the paired tests favor GHAM-2-I and LHAM-1-TIK over WS-GGI; no further paired advantage survives Holm correction. The independent BI-HMC comparisons are reported in full in Table 13: BI-HMC has significantly favorable error ranks against RLS, WS-GNI, WS-GGI, GHAM-1-dH, and GHAM-2-dH, but its comparisons with GHAM-2-I and both LHAM rules are inconclusive.

The CSTR benchmark provides stronger separation. In FSM, BI-HMC significantly outperforms all eight optimization rules. Among the optimization rules, LHAM-1-TIK and LHAM-3-TIK each significantly outperform GI and the tested GHAM variants, while neither differs significantly from Newton (Table 12). GHAM-2-dH is significantly worse than every other available method in this case. In PL, GHAM-2-I is significantly worse than all eight other available methods, while WS-GGI only establishes an advantage over GHAM-2-I. BI-HMC has favorable independent ranks against WS-GGI, GHAM-2-I, and LHAM-1-TIK, but does not establish a significant advantage over the least-squares/curvature-based group or LHAM-3-TIK. Thus neither the loss-root construction nor increased homotopy order is uniformly superior: LHAM-1-TIK outperforms LHAM-3-TIK in CSTR/FSM, whereas LHAM-3-TIK outperforms LHAM-1-TIK in CSTR/PL. These advantages also differ in practical magnitude: the median paired advantage of LHAM-1-TIK over LHAM-3-TIK in CSTR/FSM is only 1.97×10^{-4} percentage points, whereas the reverse comparison in CSTR/PL favors LHAM-3-TIK by 0.987 percentage points. BI-HMC has no significant pairwise disadvantage in the Monte Carlo cases, but the absence of separation in Example 1/FSM precludes a claim of universal superiority. The conclusions concern final parameter accuracy under the stated tuning and computational budgets. They do not establish equivalence for inconclusive pairs, asymptotic estimator consistency, or superiority in computational cost.

Table 9: Shapiro–Wilk normality diagnostics for the Monte Carlo final errors: Holm-adjusted p-values (nine testable methods per case, 30 observations each). Bold entries reject normality at 0.05.

Method	Ex. 1/FSM	Ex. 1/PL	CSTR/FSM	CSTR/PL
BI-HMC	**0.0023**	**0.0046**	**0.00807**	**0.0323**
RLS	–	$\mathbf{3.39 \times 10^{-5}}$	–	**0.00168**
Newton / WS-GNI	$\mathbf{7.86 \times 10^{-7}}$	$\mathbf{3.39 \times 10^{-5}}$	$\mathbf{2.17 \times 10^{-5}}$	**0.00168**
GI / WS-GGI	$\mathbf{7.86 \times 10^{-7}}$	$\mathbf{1.21 \times 10^{-4}}$	$\mathbf{2.17 \times 10^{-5}}$	0.396
GHAM-1-dH	$\mathbf{7.86 \times 10^{-7}}$	$\mathbf{3.39 \times 10^{-5}}$	$\mathbf{2.15 \times 10^{-5}}$	**0.00168**
GHAM-2-dH	$\mathbf{7.86 \times 10^{-7}}$	$\mathbf{3.39 \times 10^{-5}}$	$\mathbf{4.35 \times 10^{-6}}$	**0.00168**
GHAM-2-H	$\mathbf{7.86 \times 10^{-7}}$	–	$\mathbf{4.47 \times 10^{-6}}$	–
GHAM-2-I	$\mathbf{7.86 \times 10^{-7}}$	$\mathbf{3.39 \times 10^{-5}}$	$\mathbf{4.47 \times 10^{-6}}$	0.396
LHAM-1-TIK	$\mathbf{7.86 \times 10^{-7}}$	$\mathbf{9.24 \times 10^{-6}}$	$\mathbf{2.17 \times 10^{-5}}$	0.396
LHAM-3-TIK	$\mathbf{7.86 \times 10^{-7}}$	0.0837	$\mathbf{2.17 \times 10^{-5}}$	**0.016**
Normality rejected	9/9	8/9	9/9	6/9

A dash denotes an unavailable method. Newton and GI are the FSM labels corresponding to WS-GNI and WS-GGI in PL. The same BI-HMC samples are reused in the PL presentation; they are not an additional independent experiment.

Table 10: Significant pairwise differences after Holm correction: Monte Carlo realizations.

Case	Family size	Paired sign	Rank-sum
Example 1/FSM	36	0/36	–
Example 1/PL	45	2/28	5/8
CSTR/FSM	36	34/36	–
CSTR/PL	36	18/28	3/8

Entries in the last two columns are significant/testable comparisons. Both test types belong to the same Holm family within a case; studies and cases have separate families. For Example 1/PL Monte Carlo, the current export conservatively retains nine unavailable GHAM-2-H contrasts in the 45-comparison family with unit adjustment inputs; there are 36 testable contrasts. No rejection is assigned to an unavailable contrast.

Table 11: Significant advantages:disadvantages for each method in the Monte Carlo comparisons.

Method	Ex. 1/FSM	Ex. 1/PL	CSTR/FSM	CSTR/PL
BI-HMC	0:0	5:0	8:0	3:0
RLS	–	0:1	–	3:0
Newton / WS-GNI	0:0	0:1	5:1	3:0
GI / WS-GGI	0:0	0:3	4:4	1:7
GHAM-1-dH	0:0	0:1	3:5	3:0
GHAM-2-dH	0:0	0:1	0:8	3:0
GHAM-2-H	0:0	–	2:6	–
GHAM-2-I	0:0	1:0	1:7	0:8
LHAM-1-TIK	0:0	1:0	6:1	2:6
LHAM-3-TIK	0:0	0:0	5:2	3:0

Each entry counts significantly better:worse outcomes against other methods, using Holm-adjusted $p < 0.05$. These counts summarize pairwise conclusions; they are not a pooled significance test or a total ranking. A dash means there are no testable comparisons for that method.

Table 12: Selected paired exact sign tests, including inconclusive contrasts. The first-listed method is a; $D_r = E_{a,r} - E_{b,r}$. Bold p_{H} values reject equal win frequencies. All unlisted contrasts appear in the matrices.

Case	Methods a / b	$W : L : T$	p_{H}	Median D (pp)
Example 1/FSM	BI-HMC / Newton	20:10:0	1	-0.224
Example 1/PL	GHAM-2-I / WS-GGI	28:2:0	$\mathbf{3.91 \times 10^{-5}}$	-0.692
Example 1/PL	LHAM-1-TIK / WS-GGI	25:5:0	**0.014**	-0.607
CSTR/FSM	BI-HMC / Newton	29:1:0	$\mathbf{1.79 \times 10^{-6}}$	-0.449
CSTR/FSM	LHAM-1-TIK / GI	29:1:0	$\mathbf{1.79 \times 10^{-6}}$	-0.00499
CSTR/FSM	LHAM-3-TIK / GI	29:1:0	$\mathbf{1.79 \times 10^{-6}}$	-0.00471
CSTR/FSM	LHAM-1-TIK / Newton	17:13:0	0.585	-2.64×10^{-5}
CSTR/FSM	LHAM-3-TIK / Newton	11:19:0	0.401	1.14×10^{-4}
CSTR/FSM	LHAM-1-TIK / LHAM-3-TIK	23:7:0	**0.0157**	-1.97×10^{-4}
CSTR/PL	LHAM-3-TIK / LHAM-1-TIK	27:3:0	$\mathbf{1.69 \times 10^{-4}}$	-0.987
CSTR/PL	WS-GGI / GHAM-2-I	30:0:0	$\mathbf{6.33 \times 10^{-8}}$	-5.34

All rows contain 30 jointly valid records. Negative median differences favor the first method; pp denotes percentage points of relative parameter error. Effect sizes are descriptive and do not turn the sign test into a test of means.

Table 13: Independent comparisons of BI-HMC against the PL methods. $\widehat{A}_{\mathrm{BI,a}}$ estimates $P(E_{\mathrm{BI}} < E_a) + \frac{1}{2}P(E_{\mathrm{BI}} = E_a)$; values above 0.5 favor BI-HMC. Bold p_{H} values reject equal distributions.

PL comparator	Ex. 1 $\widehat{A}$	Ex. 1 p_{H}	CSTR $\widehat{A}$	CSTR p_{H}
RLS	0.748	**0.0421**	0.653	0.421
WS-GNI	0.748	**0.0421**	0.653	0.421
WS-GGI	0.860	$\mathbf{7.61 \times 10^{-5}}$	1.000	$\mathbf{1.09 \times 10^{-9}}$
GHAM-1-dH	0.748	**0.0421**	0.653	0.421
GHAM-2-dH	0.748	**0.0421**	0.653	0.421
GHAM-2-I	0.650	1	1.000	$\mathbf{1.09 \times 10^{-9}}$
LHAM-1-TIK	0.672	0.738	0.886	$\mathbf{6.32 \times 10^{-6}}$
LHAM-3-TIK	0.712	0.185	0.706	0.0957

Each test uses 30 independent final estimates from each method. The 900 cross-sample pairs estimate A but are not 900 independent trials. These are two-sided Wilcoxon–Mann–Whitney tests; no binomial test is applied to the cross-sample pair counts.

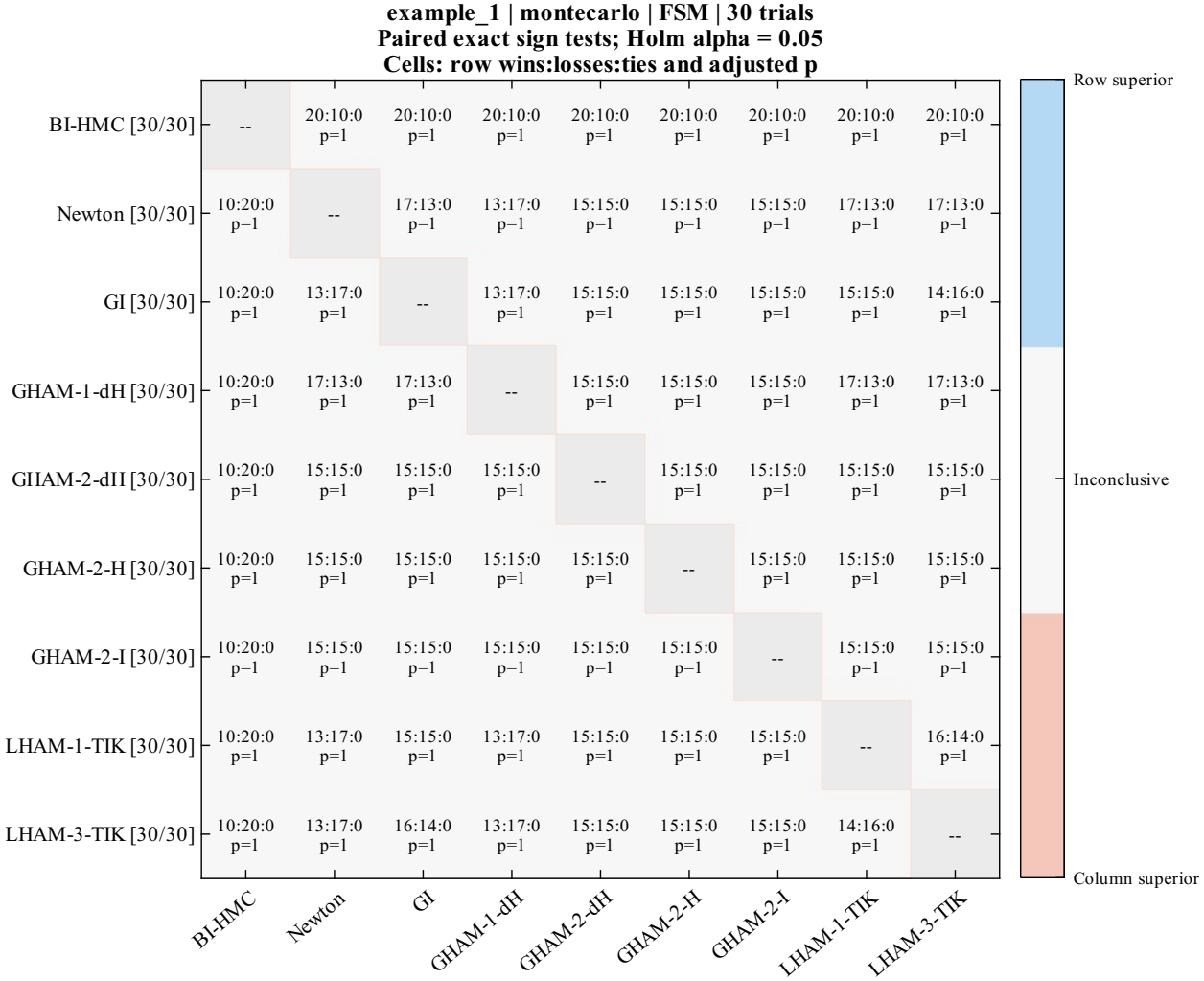


Figure 19: Final-parameter-error comparison matrix for Example 1/FSM, 30 independent Monte Carlo records. Blue cells favor the row method and red cells favor the column method after Holm correction ($\alpha = 0.05$); white cells are inconclusive. Paired cells show row wins:losses:ties and p_{H}. Cells marked RS use independent rank-sum tests and show $\widehat{A}_{\mathrm{row,col}}$ and p_{H}. Grey diagonal or NA entries are not tests.

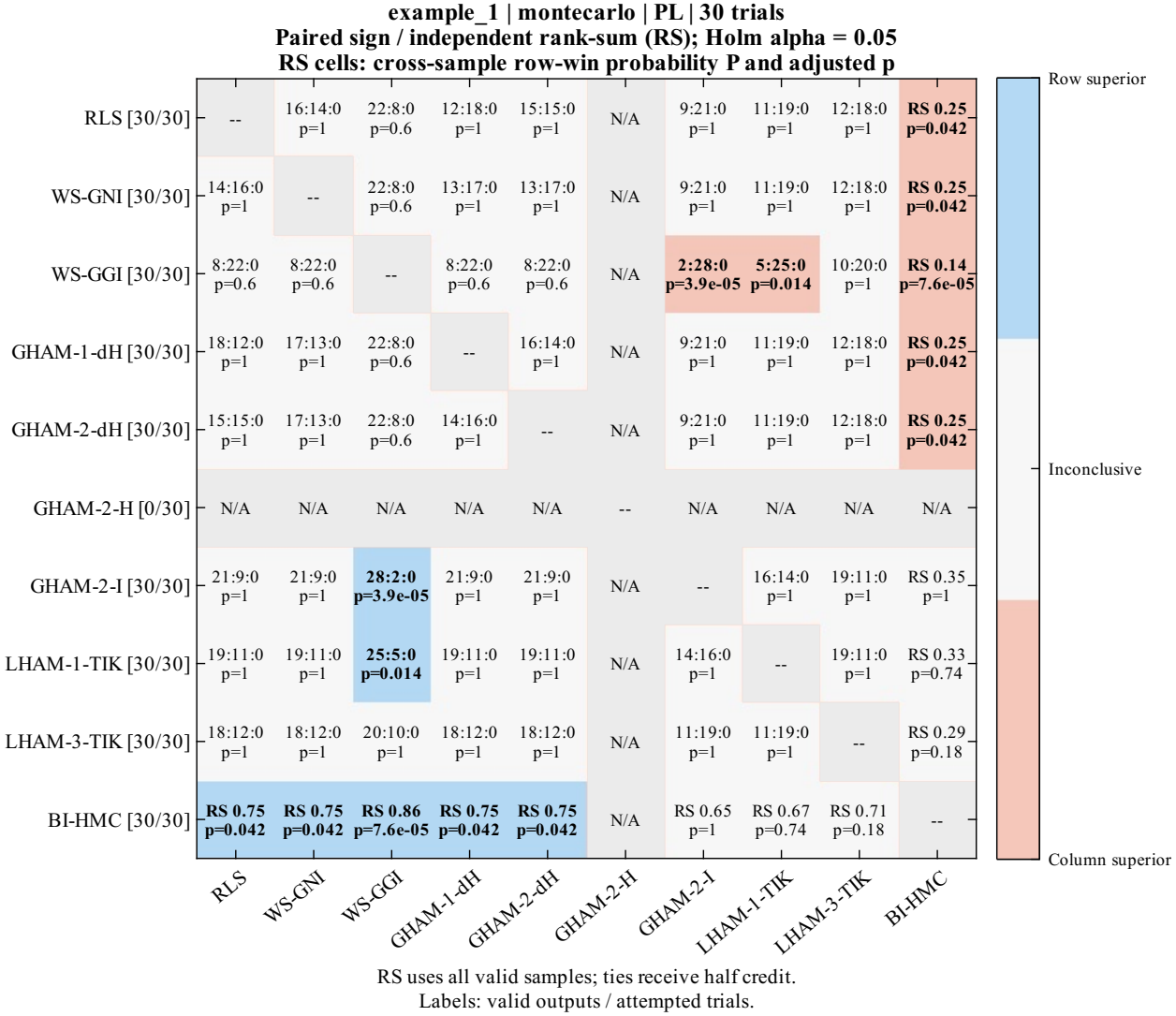


Figure 20: Final-parameter-error comparison matrix for Example 1/PL, 30 independent Monte Carlo records. Blue cells favor the row method and red cells favor the column method after Holm correction ($\alpha = 0.05$); white cells are inconclusive. Paired cells show row wins:losses:ties and p_{H}. Cells marked RS use independent rank-sum tests and show $\widehat{A}_{\mathrm{row,col}}$ and p_{H}. Grey diagonal or NA entries are not tests.

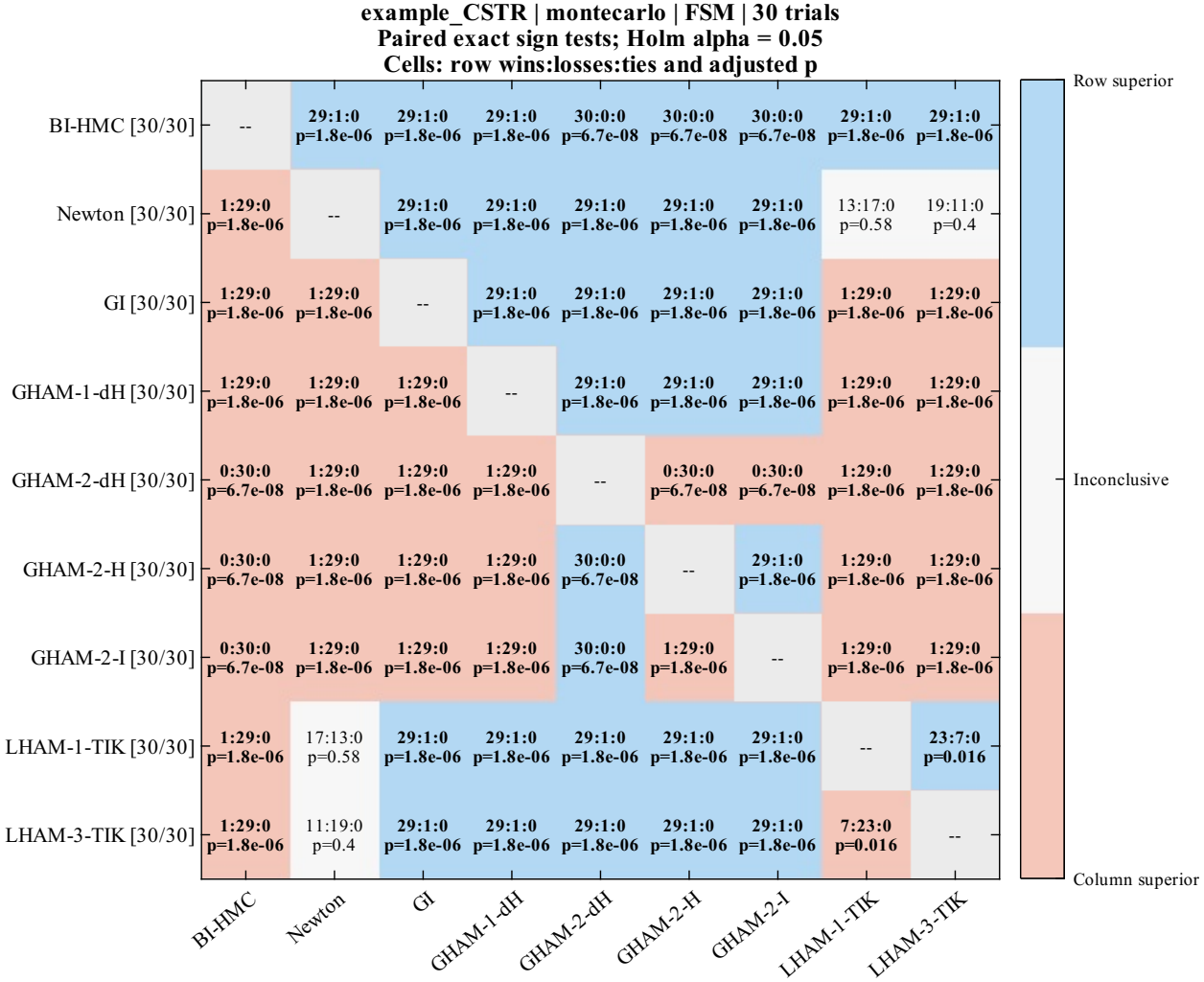


Figure 21: Final-parameter-error comparison matrix for CSTR/FSM, 30 independent Monte Carlo records. Blue cells favor the row method and red cells favor the column method after Holm correction ($\alpha = 0.05$); white cells are inconclusive. Paired cells show row wins:losses:ties and p_{H}. Cells marked RS use independent rank-sum tests and show $\widehat{A}_{\mathrm{row,col}}$ and p_{H}. Grey diagonal or NA entries are not tests.

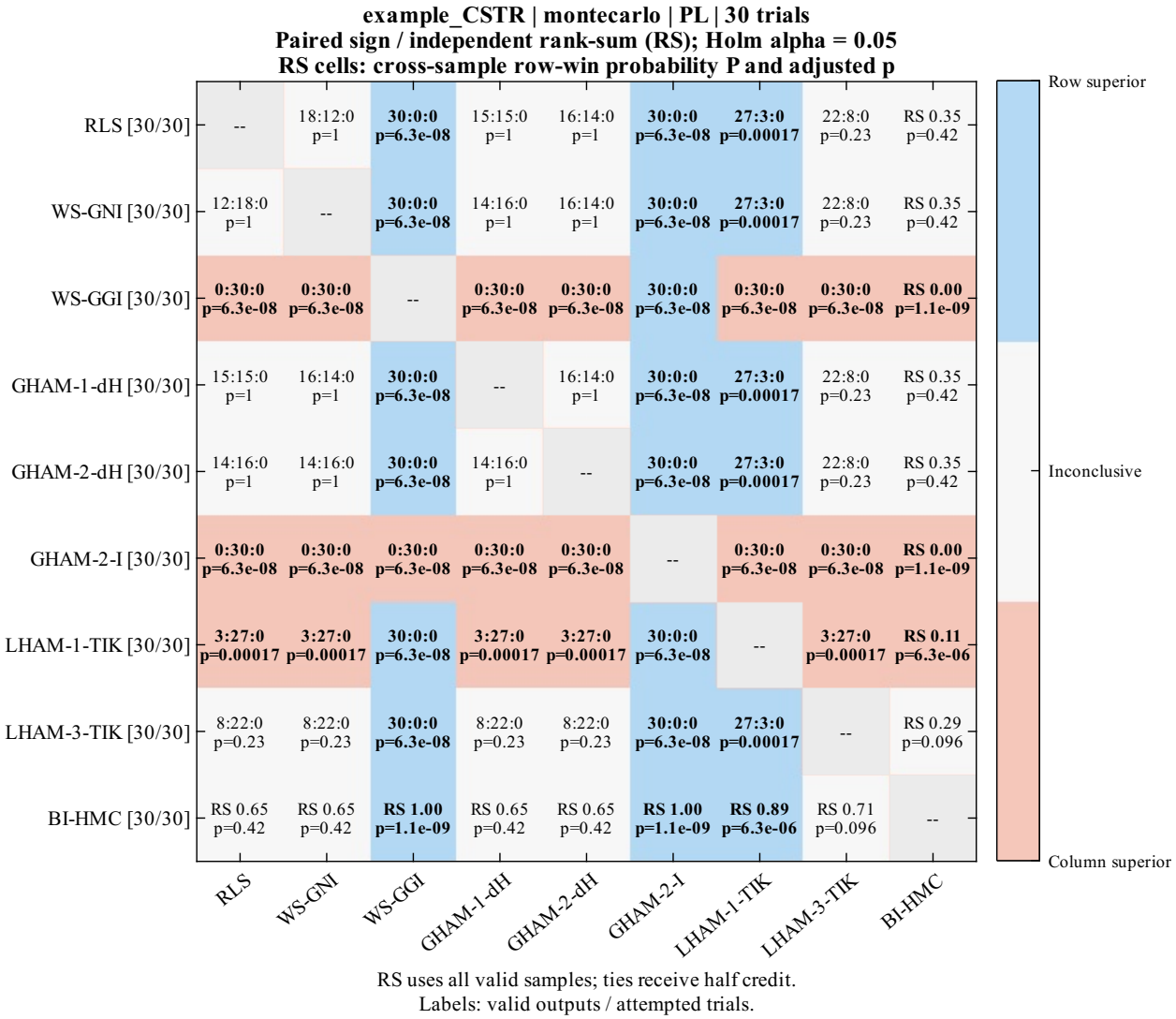


Figure 22: Final-parameter-error comparison matrix for CSTR/PL, 30 independent Monte Carlo records. Blue cells favor the row method and red cells favor the column method after Holm correction ($\alpha = 0.05$); white cells are inconclusive. Paired cells show row wins:losses:ties and p_{H}. Cells marked RS use independent rank-sum tests and show $\widehat{A}_{\mathrm{row,col}}$ and p_{H}. Grey diagonal or NA entries are not tests.

5.5 Robustness to parameter initialization

The initialization ablation varies the starting point over 30 independently drawn initializations while retaining a fixed input/noise record within each implementation. Final errors were compared using the outcome and tests in Section 5.4, with separate Holm families for this study. Paired tests use the shared starting point and data record of each original experiment. When a method fails to return a valid estimate, its output cannot define a parameter-error difference; accuracy tests therefore condition on jointly valid pairs and failed outputs are not awarded as wins to their competitors. All finite extreme outcomes were retained, including those suppressed only in boxplot displays.

Tables 14– 16 and Figures 23– 26 show strong initialization-dependent differences. BI-HMC provides valid point estimates for all 30 initializations and has significantly favorable errors against all eight available comparators in each reported case. The PL BI-HMC contrasts require particular qualification: the reused Python reference and the MATLAB methods use different fixed data records and different starting-point draws. Their rank-sum results compare these conditional error distributions; they do not isolate the effect of initialization or establish a causal robustness advantage under identical observed data. The FSM paired contrasts provide the directly controlled comparison.

Among the optimization rules, LHAM-3-TIK is favorable in Example 1/PL, with significant advantages over six comparators and 30 valid outcomes; its comparison with LHAM-1-TIK is inconclusive. This behavior does not transfer uniformly to CSTR/PL, where RLS, WS-GNI, GHAM-1-dH, and GHAM-2-dH attain significantly lower errors than both LHAM variants on jointly valid starts. In the two PL cases, however, RLS and WS-GNI return only 14 and 16 finite estimates, respectively, whereas both LHAM variants return 30. Consequently, conditional accuracy and successful completion must be considered separately. For Example 1/FSM, GHAM-2-H and GHAM-2-I frequently produce smaller errors than the LHAM rules, but GHAM-2-I has only 18 valid outcomes and GHAM-2-H includes a rare, very large finite error. A sign-test advantage alone does not imply smaller tail risk. For CSTR/FSM, both LHAM variants significantly outperform all evaluated GHAM rules but neither significantly outperforms Newton or GI. The initialization study therefore supports case-specific trade-offs between conditional accuracy and completion frequency, rather than a single optimizer that is uniformly robust to initialization.

Table 14: Significant pairwise differences after Holm correction: initialization ablation.

Case	Family size	Paired sign	Rank-sum
Example 1/FSM	36	26/36	–
Example 1/PL	36	14/28	8/8
CSTR/FSM	36	30/36	–
CSTR/PL	36	24/28	8/8

The paired accuracy tests condition on jointly valid outcomes. Independent RS tests involving the reused BI-HMC reference compare separate starting-point samples on different fixed records; they do not isolate initialization robustness under identical observed data.

Table 15: Valid final estimates among 30 attempted initializations.

Method	Ex. 1/FSM	Ex. 1/PL	CSTR/FSM	CSTR/PL
BI-HMC	30	30	30	30
RLS	–	14	–	14
Newton / WS-GNI	30	16	30	16
GI / WS-GGI	24	30	30	30
GHAM-1-dH	30	29	30	30
GHAM-2-dH	28	26	30	28
GHAM-2-H	30	–	30	–
GHAM-2-I	18	28	30	28
LHAM-1-TIK	30	30	30	30
LHAM-3-TIK	30	30	30	30

A dash denotes an unavailable method. Finite estimates are not necessarily converged; the saved valid-output policy also includes finite stalled or iteration-limited estimates and valid Bayesian point estimates with sampler warnings. No success-rate hypothesis is inferred from the accuracy tests.

Table 16: Significant advantages:disadvantages for each method in the initialization comparisons.

Method	Ex. 1/FSM	Ex. 1/PL	CSTR/FSM	CSTR/PL
BI-HMC	8:0	8:0	8:0	8:0
RLS	–	1:2	–	5:1
Newton / WS-GNI	3:4	1:2	4:1	4:1
GI / WS-GGI	0:5	0:7	4:1	1:7
GHAM-1-dH	4:3	2:2	3:5	4:3
GHAM-2-dH	0:1	2:2	0:8	5:1
GHAM-2-H	5:1	–	1:6	–
GHAM-2-I	5:1	0:5	1:6	0:8
LHAM-1-TIK	1:5	2:1	5:1	2:6
LHAM-3-TIK	0:6	6:1	4:2	3:5

Each entry counts significantly better:worse outcomes against other methods, using Holm-adjusted $p < 0.05$. These counts summarize pairwise conclusions; they are not a pooled significance test or a total ranking. A dash means there are no testable comparisons for that method.

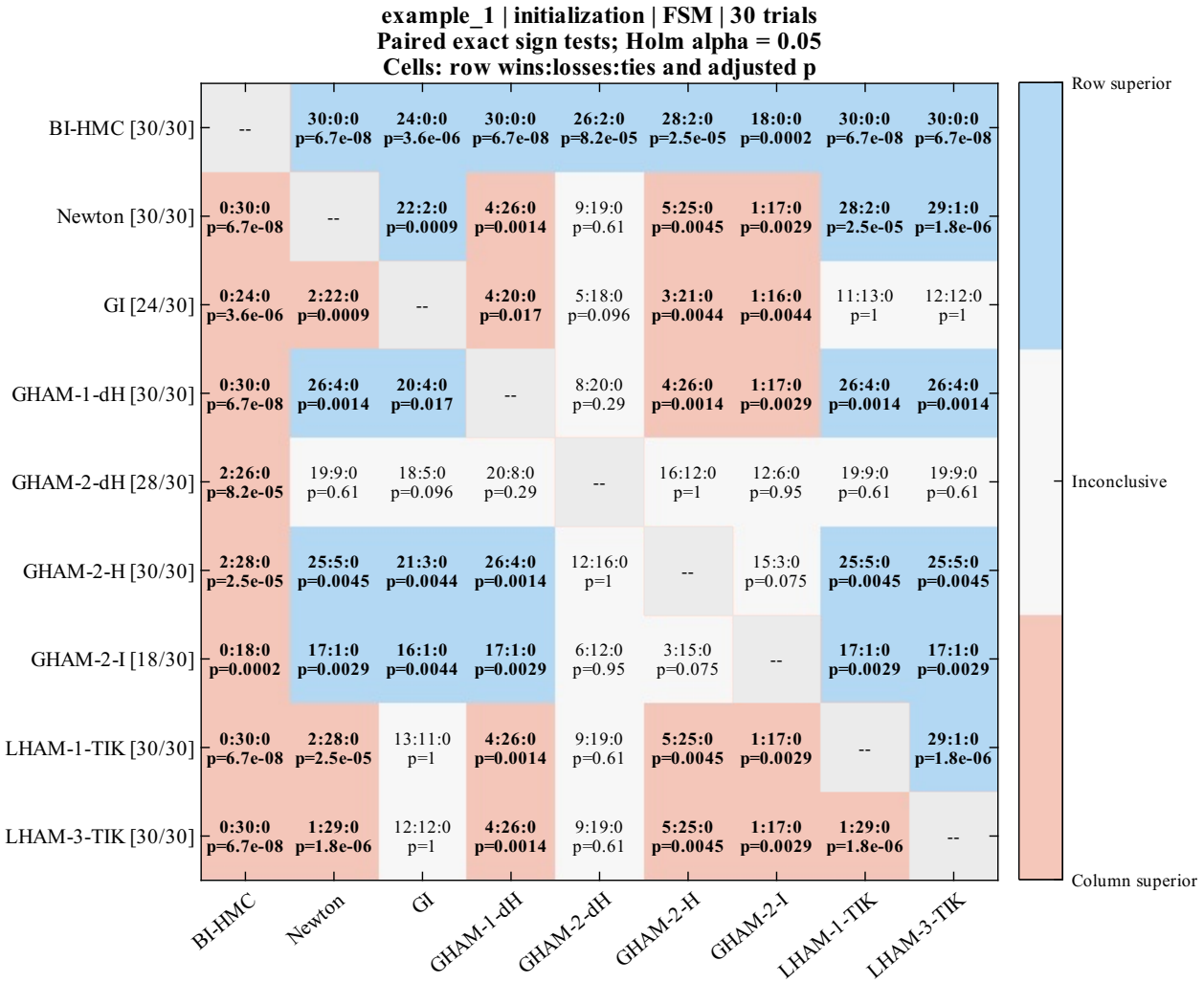


Figure 23: Final-parameter-error comparison matrix for Example 1/FSM, 30 randomized initializations. Blue cells favor the row method and red cells favor the column method after Holm correction ($\alpha = 0.05$); white cells are inconclusive. Paired cells show row wins:losses:ties and p_{H}. Cells marked RS use independent rank-sum tests and show $\widehat{A}_{\mathrm{row,col}}$ and p_{H}. Grey diagonal or NA entries are not tests.

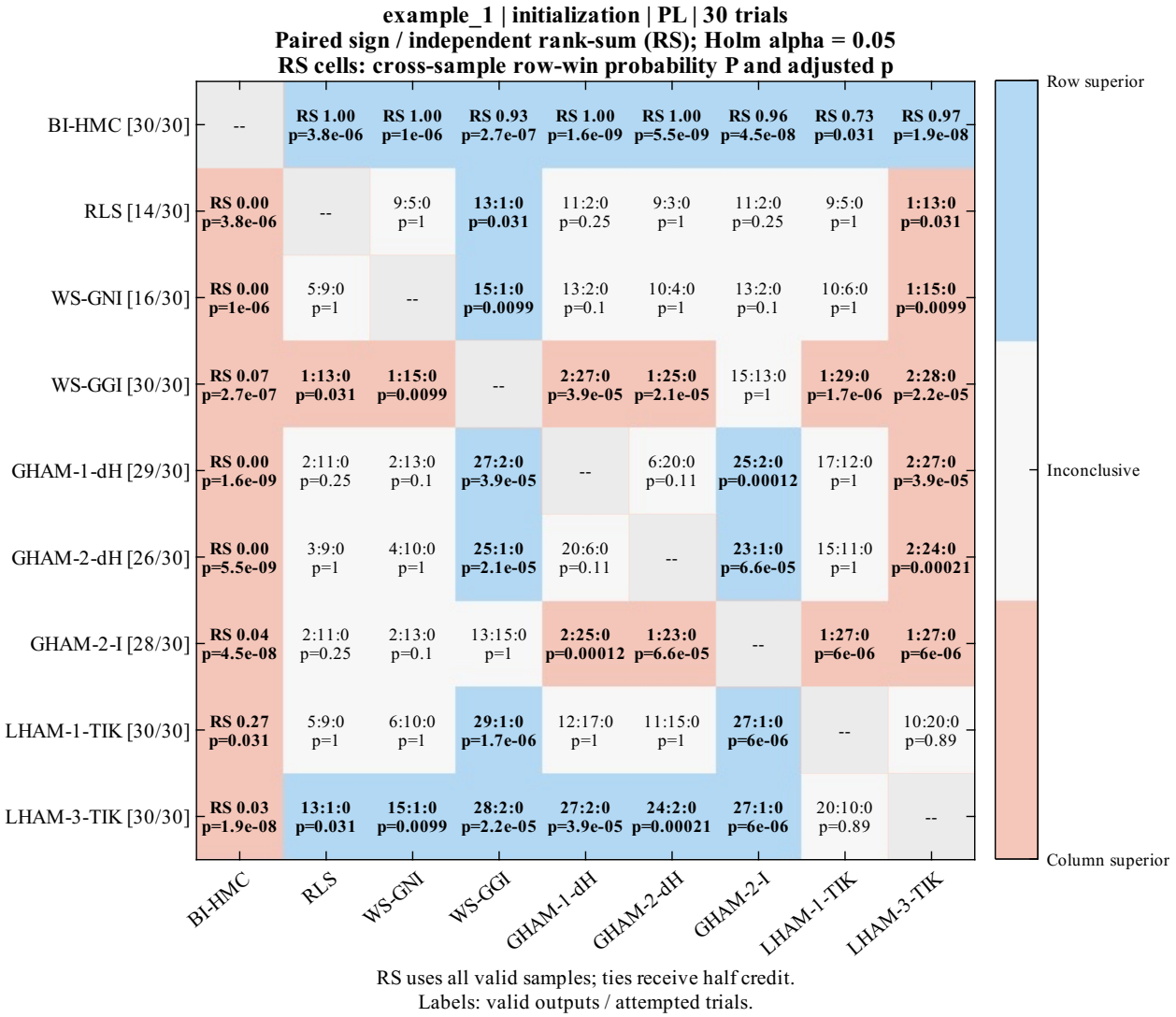


Figure 24: Final-parameter-error comparison matrix for Example 1/PL, 30 randomized initializations. Blue cells favor the row method and red cells favor the column method after Holm correction ($\alpha = 0.05$); white cells are inconclusive. Paired cells show row wins:losses:ties and p_{H}. Cells marked RS use independent rank-sum tests and show $\widehat{A}_{\mathrm{row,col}}$ and p_{H}. Grey diagonal or NA entries are not tests.

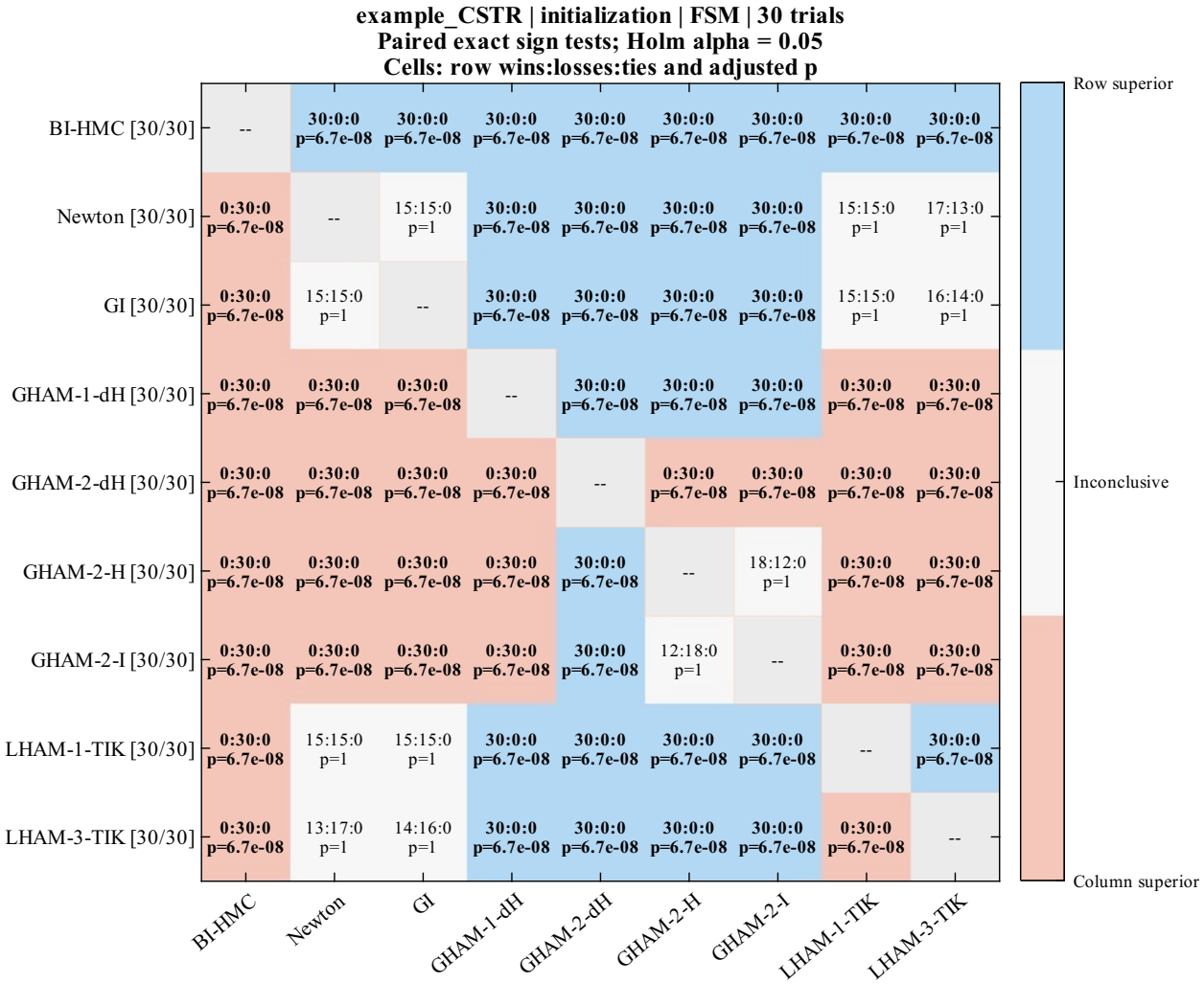


Figure 25: Final-parameter-error comparison matrix for CSTR/FSM, 30 randomized initializations. Blue cells favor the row method and red cells favor the column method after Holm correction ($\alpha = 0.05$); white cells are inconclusive. Paired cells show row wins:losses:ties and p_{H}. Cells marked RS use independent rank-sum tests and show $\widehat{A}_{\mathrm{row,col}}$ and p_{H}. Grey diagonal or NA entries are not tests.

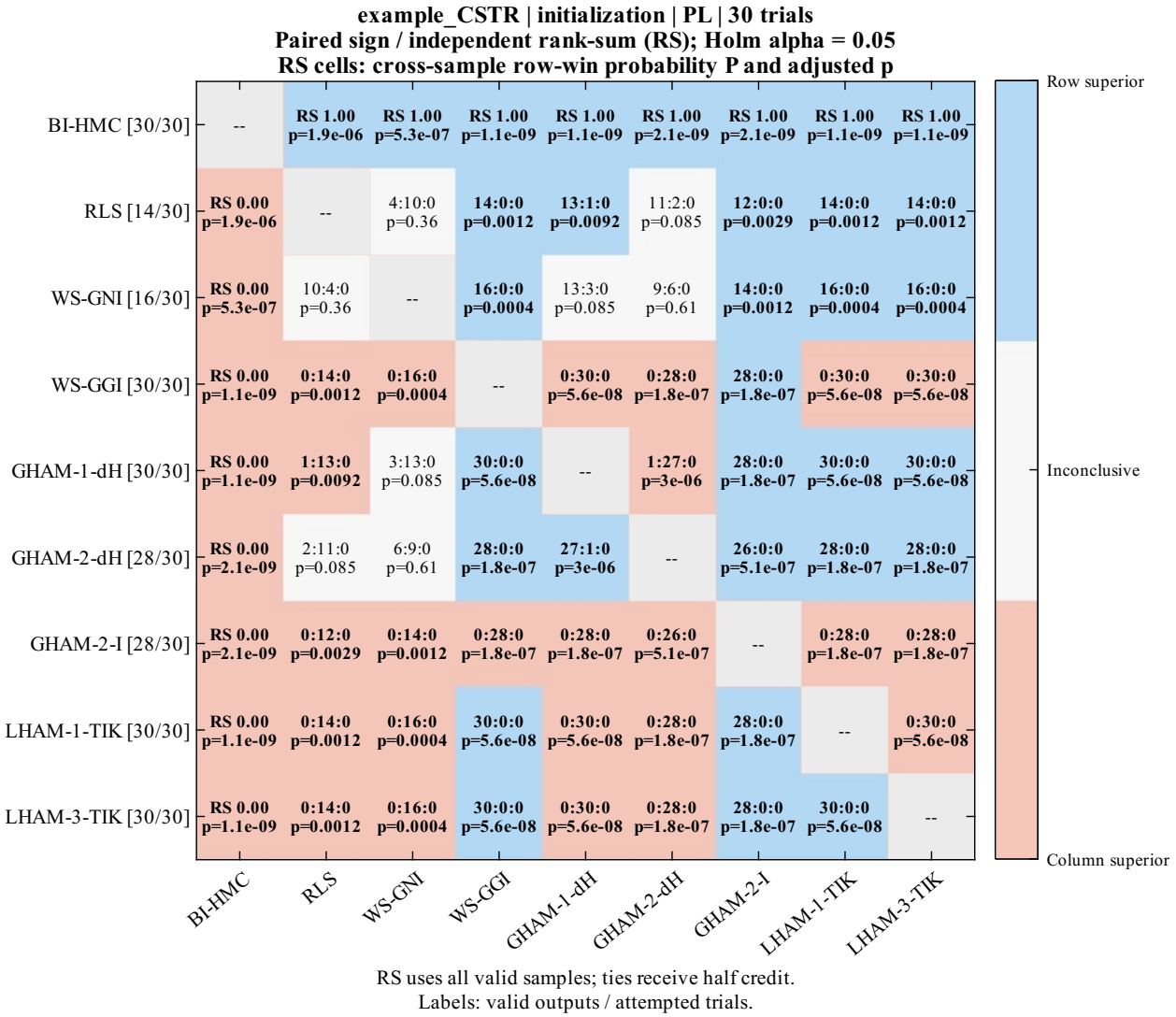


Figure 26: Final-parameter-error comparison matrix for CSTR/PL, 30 randomized initializations. Blue cells favor the row method and red cells favor the column method after Holm correction ($\alpha = 0.05$); white cells are inconclusive. Paired cells show row wins:losses:ties and p_{H}. Cells marked RS use independent rank-sum tests and show $\widehat{A}_{\mathrm{row,col}}$ and p_{H}. Grey diagonal or NA entries are not tests.

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